

* stabilizer codes

→ Definition

- Stabilizer generators all commute with each other

$$P_j P_k = P_k P_j \quad \text{for all } j, k \in \{1, \dots, r\}$$

- They form a minimal generating set

$$P_k \notin \langle P_1, \dots, P_{k-1}, P_{k+1}, \dots, P_r \rangle$$

↳ Removing any P_a , must change the set size
↓

No generator is non-essential

- At least one quantum state vector is fixed by all stabilizer generators

$$P_i | \psi \rangle = | \psi \rangle \quad \text{for all } i \in \{1, \dots, r\} \quad \rightarrow | \psi \rangle \text{ exists}$$

Equivalently,

$$-I^n \notin \langle P_1, \dots, P_r \rangle$$

$$HXH^\dagger = Z$$

$$HYH^\dagger = -Y$$

$$HZH^\dagger = X$$

$$SX S^\dagger = -Y$$

$$SY S^\dagger = -X$$

$$SZ S^\dagger = Z$$

→ Quantum computations involving Clifford operations only can easily be simulated using classical computation

↓
Gottesman-Knill theorem

* Encoders for stabilizer codes

→ Any stabilizer code that encodes m qubits into n qubits, there's always a Clifford operation U such that

$$| \psi \rangle = U (| \phi \rangle \otimes | 0^{n-m} \rangle)$$

$| \phi \rangle = m$ qubits

$| \psi \rangle = n$ qubits

↳ The encoding requires at max
 $O\left(\frac{n^2}{\log(n)}\right)$ gates

↓
Optimization technique can be found
in proof of Gottesman-Knill Theorem

* Dimensions of any subspace is equal to the trace of product of projections onto that subspace

$$\Pi_{1, \dots, r} \cdot \Pi_r = \frac{1}{2^n} \sum_{a_i \in \{0,1\}} P_{1, \dots, r}^{a_i} \cdot P_r^{a_i}$$

$i \in \{1, \dots, r\}$

$$\therefore \dim(C) = \text{Tr}(\Pi_{1, \dots, r} \cdot \Pi_r) = \frac{2^n}{2^r} = 2^{n-r}$$

$$\text{Tr}(\mathbb{I}^{2^n}) = 2^n, \quad \text{Tr}(P_{1, \dots, r}^{a_i} \cdot P_r^{a_i}) = 0$$

for $P_{1, \dots, r}^{a_i} \cdot P_r^{a_i} \neq \mathbb{I}^{2^n}$

Ex. * Projections on 2-D Hilbert space $H = \text{span}\{|0\rangle, |1\rangle\}$

$$P = |0\rangle\langle 0| + |1\rangle\langle 1| \quad \text{Tr}(P) = 2 = \dim(H)$$

* Clifford Operations

→ Unitary operations on any no. of qubits can be implemented by using,

Hadamard Gates

S Gates

CNOT Gates

→ Clifford operations are operations that transform Pauli operations into Pauli operations upto a global phase factor

$$UPU^\dagger = \pm Q$$

→ $P, Q \rightarrow$ Pauli operation,

→ $U \rightarrow$ Clifford operation

→ GHZ Stabilizer code

$$I \quad I \quad Z \quad Z \quad I$$

$$I \quad I \quad I \quad Z \quad Z$$

$$I \quad I \quad X \quad X \quad X$$

* Code Space Dimensions

Code space defined by $\langle P_1, \dots, P_r \rangle$ has 2^{n-r} dimensions

Intuitively, each stabilizer generator takes away 1 of the n qubits, in other words, each stabilizer generator divides the dimensions in half

Every element of the stabilizer can be written as

$$P_1^{a_1} \dots P_r^{a_r} \quad a_1, \dots, a_r \in \{0, 1\}$$

$$\Pi_k = \frac{I^{\otimes n} + P_k}{2} \quad \rightarrow \text{Projection onto the space of } r+1 \text{ eigenvectors of } P_k$$

$$k \in \{1, \dots, r\}$$

$$P_k | \psi \rangle = | \psi \rangle \quad \rightarrow \quad \Pi_k | \psi \rangle = | \psi \rangle$$

$$P_k | \psi \rangle = - | \psi \rangle \quad \rightarrow \quad \Pi_k | \psi \rangle = 0$$

$\Pi_1, \dots, \Pi_r$ - Projection onto Code Space C

→ 9-qubit Shor code

Z	Z	I	I	I	I	I	I	I
I	Z	Z	I	I	I	I	I	I
I	I	I	Z	Z	I	I	I	I
I	I	I	I	Z	Z	I	I	I
I	I	I	I	I	I	Z	Z	I
I	I	I	I	I	I	I	Z	Z
X	X	X	X	X	X	I	I	I
I	I	I	X	X	X	X	X	X

→ 7-qubit Steane code

Z	Z	Z	Z	I	I	I
Z	Z	I	I	Z	Z	I
Z	I	Z	I	Z	I	Z
X	X	X	X	I	I	I
X	X	I	I	X	X	I
X	I	X	I	X	I	X

→ 5-qubit code

X	Z	Z	X	I
I	X	Z	Z	X
X	I	X	Z	Z
Z	X	I	X	Z

→ 6-bit stabilizer code

Z	Z
X	X

Code Space Definition

$$C = \{ \mathbf{c} : P_i \mathbf{c} = \mathbf{c}_i \text{ for all } i \in \{1, \dots, r\} \}$$

$$\downarrow \qquad \qquad \qquad \downarrow$$

Code Space In $(+1, +1, +1, \dots, +1, +1)$

Examples

3-bit repetition code

$$(2 \otimes 2 \otimes 1) (1 \otimes 2 \otimes 2) : (1 \otimes 2 \otimes 2) (2 \otimes 2 \otimes 1)$$

$$2 \otimes 2 \otimes 1 \notin \langle 1 \otimes 2 \otimes 2 \rangle : \{ 1 \otimes 1 \otimes 1, 1 \otimes 2 \otimes 2 \}$$

$$1 \otimes 2 \otimes 2 \notin \langle 2 \otimes 2 \otimes 1 \rangle : \{ 1 \otimes 1 \otimes 1, 2 \otimes 2 \otimes 1 \}$$

$$- 1 \otimes 1 \otimes 1 \notin \langle 2 \otimes 2 \otimes 1, 1 \otimes 2 \otimes 2 \rangle$$

modified 3-bit repetition code

$$X \otimes X \otimes I$$

$$I \otimes X \otimes X$$

Conventionally written as

$$X \otimes X \otimes I$$

$$I \otimes X \otimes X$$

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$\therefore \text{max. weight of } C = \frac{d-1}{2}$

$$P_k E = \alpha E P_k \quad P_k C = \alpha C P_k \quad \alpha \in \{+1, -1\}$$

$\hookrightarrow E, C$ must commute or anti-commute with all stabilizer generators

$$P_k (CE) = \alpha C P_k E = \alpha^2 CE P_k = (CE) P_k$$

$\hookrightarrow CE$ commutes with all stabilizer generators

$\downarrow$
It is proportional to the stabilizer elements

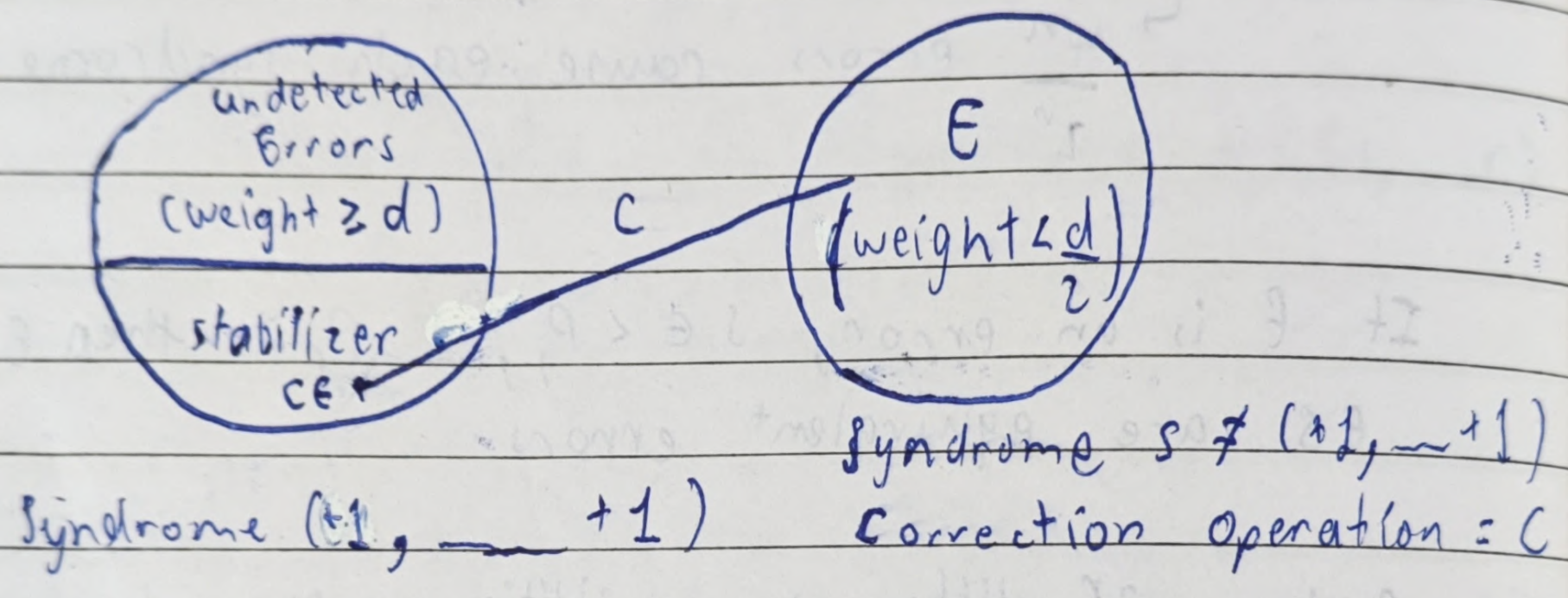
$\rightarrow$ It is however computationally difficult to compute the lowest weight Pauli operation causing a given syndrome

→ Natural strategy → Choose lowest weight Pauli operation that, as an error, causes the syndrome

↓
Lower weight Pauli operations represent more likely explanations for the syndrome measured

→ For a distance d stabilizer code the correction operation should weigh at most $\frac{(d-1)}{2}$ as per

the natural strategy



weight of $CE < d$

weight of $C < \text{weight of } E$

↳ C has minimal weight

⇒ weight of $E < \frac{d}{2}$

⇒ weight of $C < \text{weight of } E < \frac{d}{2}$

⇒ weight of $C < \frac{d}{2}$

$$\begin{array}{cccccccccccc} I & \otimes & I & \otimes & I & \otimes & I & \otimes & X & \otimes & X & \otimes & X \\ I & \otimes & I & \otimes & I & \otimes & I & \otimes & Z & \otimes & Z & \otimes & Z \end{array}$$

$\left. \begin{array}{l} \text{Errors go undetected,} \\ \text{make non-trivial change,} \\ \text{aren't proportional to} \\ \text{stabilizer elements} \end{array} \right\}$

* Correcting Errors

2^r distinct syndromes divide 4^n Pauli operations in equal sized collections

$\hookrightarrow \frac{4^n}{2^r}$ errors cause each syndrome

If E is an error, $S \in \langle P_1, \dots, P_r \rangle$ then E and ES are equivalent errors

S has 2^r different possibilities $\rightarrow$ No. of elements in $\langle P_1, \dots, P_r \rangle$

This leaves $\frac{4^n}{2^r} \cdot \frac{1}{2^r} = 4^{n-r}$ inequivalent classes of errors

$\downarrow$
unless $n=r$, all errors can't be corrected

$\rightarrow$ we must choose one correction operation for each syndrome

→ Case-2

$E \neq \pm Q$ for $Q \in \langle P_1, \dots, P_r \rangle$ but $EP_k = P_k E$
for every $k \in \{1, \dots, r\}$

→ It changes the code space in some non-trivial way but syndrome is $(+1, \dots, +1)$ so it remains undetected

→ Ex. → $X \otimes X \otimes X$ error on $\alpha|000\rangle + \beta|111\rangle$

→ Case-3

$EP_k = -P_k E$ for at least one $k \in \{1, \dots, r\}$

→ Error is detected by the code

→ Distance of a Stabilizer code

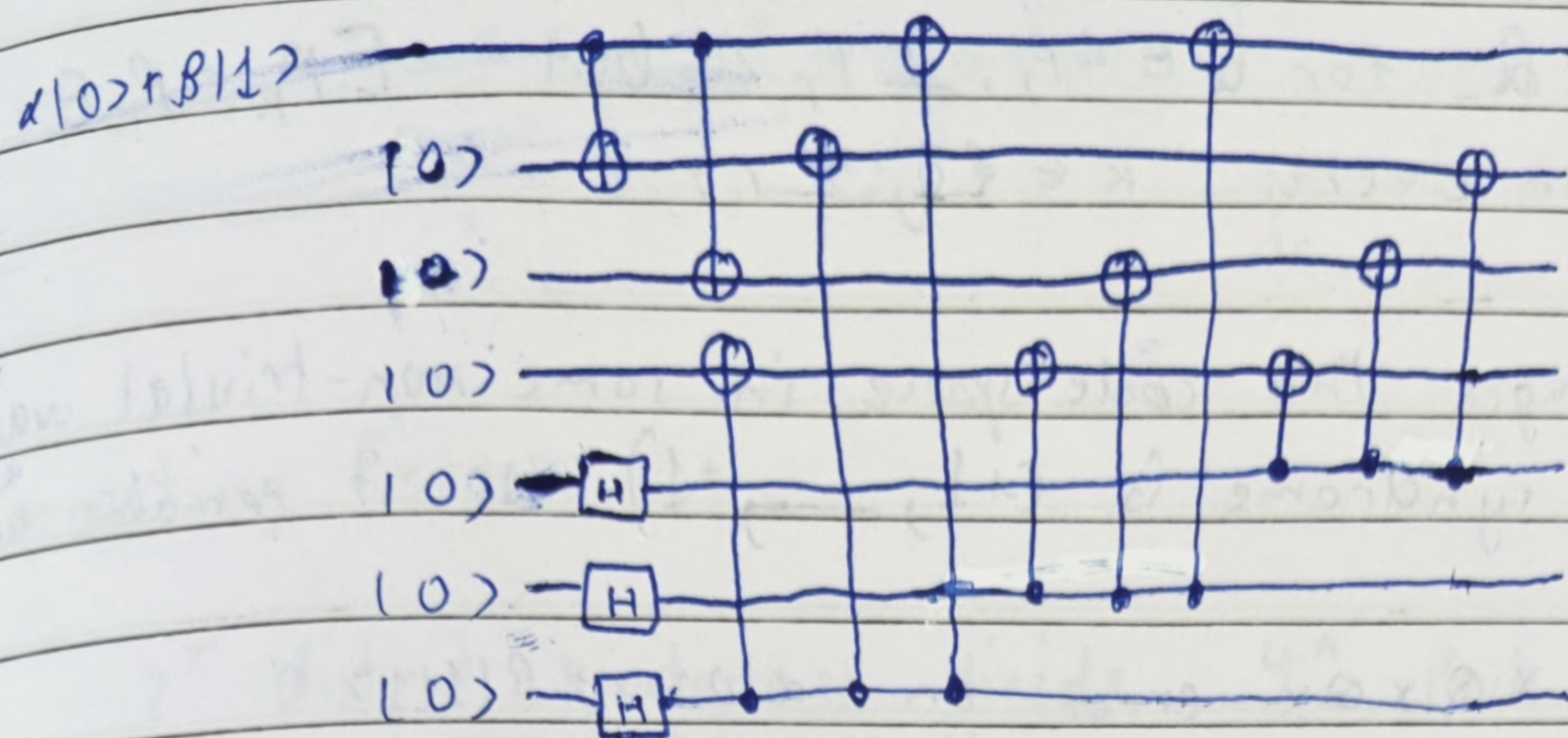
→ Distance is the minimum weight of an n -qubit Pauli operation that falls in Case-2, which means, it changes code space in some non-trivial way but goes undetected

→ $[[n, m, d]]$ stabilizer code → encodes m qubits into n qubits, has distance d

→ Ex. → 7-qubit Steane code → Distance = 3

↳ can be evaluated by considering an arbitrary 7-qubit Pauli operation with weight 2

Ex. → 7-qubit Steane code



* Detecting Errors

→ Stabilizer generators are measured as observables to detect errors

→ The r outcomes, each one $+1$ or -1 , help us detect error

→ Error Detection Cases

→ Suppose there's an error $E \rightarrow$ an n -qubit Pauli operation

→ Case-1

$$E \in \mathcal{Q} \quad \text{for } \mathcal{Q} \in \langle P_1, \dots, P_r \rangle$$

→ E commutes with all stabilizer generators, it isn't detected as an error

→ It only changes $|1\rangle$ upto a global phase, so we can ignore it